

SE322: Software Design and Architecture

Activity 4: UMLII

Objective: In this activity, the students will learn how draw activity diagram, and sequence diagram.

Expected Outcome: After completing this exercise, the students will be able to draw:

1. Activity diagram
2. Sequence diagram

Course Learning Outcome:

1. CLO 3: Differentiate how various architectural styles and design patterns enhance and degrade a system's functional-and non-functional properties.
2. CLO 5: Present concise and unambiguous architecture and design descriptions.
3. CLO 6: Create, implement, and present an architecture and design in a group project.

Tool needed:

Visio, Lucidchart or StarUml

Resource:

1. Use case scenarios
http://www.utm.mx/~caff/doc/OpenUPWeb/openup/guidances/guidelines/detail_ucs_and_scenarios_6BC56BB7.html
2. Activity diagram
<http://agilemodeling.com/artifacts/activityDiagram.htm>
3. Sequence diagram
<http://www.agilemodeling.com/artifacts/sequenceDiagram.htm>

Question1 :

The ATM System first displays menu of withdrawal amount and option to cancel, if the user cancels the transaction, the menu program exits. The Customer selects enter the amount he wishes to withdraw If the required amount is greater than the available balance, it displays an appropriate error message. Otherwise it tests whether sufficient cash is available in the cash dispenser. If there is insufficient cash in the cash dispenser, it displays an appropriate error message. The ATM system it interacts with the database to debit amount from user's account. Then it dispenses cash and instructs the user to take cash.

1. Prepare a Use Case scenario to document the Withdraw cash use case in the ATM System as described above
2. Draw Activity diagram for the Withdraw cash in the ATM System as described above
3. Create sequence diagram for the Withdraw cash in the ATM System as described above



